Supplementary Document: Testing the Promises of Blockchain-Based Consent Withdrawal in Multi-Hospital Federated Learning: An Empirical Study With Electrocardiogram Data

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Extended results (Tables R1–R23, Figs. R1–R5), including the v3.2 contract details, implementation notes, training and statistical settings, and cost measurements, are in the code repository (https://github.com/WooyeongJang/consent-withdrawal-fl, `docs/extended_results.pdf`). Tables are numbered in the order of their first citation in the main text.

S1. Evidence Behind Each Conclusion

Table S1 maps each conclusion of the main text to the kind of evidence behind it: measured, design analysis, local unit tests, or not evaluated. A re-derivation by the authors from ledger events, stored checkpoints, and run statistics, independent of the experiment's audit code but not a third-party reproduction, matched 91 of 101 claims; 3 mismatches were errors of statement, corrected here, and 7 rested only on the experiment's own records: the use of the withdrawn records in the 16 unsampled T2b runs, T5's L1 and L2 results, 95 of the 100 T6 detections, T9b and T9c (by construction), the B6 column (by definition), and the L1 re-executions, which have since been repeated by the authors for all 1,083 honest units (877 type-A, 206 type-B) from the ledger's sampling seeds and commitments, in the pinned environment, and all matched (extended results, Section R12).

TABLE S1
Evidence behind each conclusion of the main text: measured, design analysis, local unit tests, or not evaluated.

Conclusion	Evidence
Selection-time checks miss withdrawals during training; acceptance-time checks do not (C1)	measured: E1 simulation (20 seeds, seven conditions) and Kairos (six rounds); ledger race: v3.2 row below
Blocking is not removal after late withdrawal; L3 lacks power after early or mid withdrawal and lacks precision at 1% (C2)	measured: E2 (19 scenarios × 10 seeds), paired tests, TOST with positive control; outcomes recomputed under the rule as written (Section S6)
Recording-only designs cannot observe remediation-stage faults and concealment cases (C3)	observed: completion records written on the ledger for 5 honest and 510 injected jobs; design analysis for the verdicts of B0–B3 and B5; measured for B4 and B6 (recomputed)
B4 detects injected faults and concealment cases (12 of 13 types in every injection)	measured, for faults and concealment cases we designed against B4 (E3, including T11); T1, T9b, T9c by construction
v3.2 prevents the ledger race and most v3.1 gaps (four remain)	race: Kairos, two concurrent submitters (v3.1: 54 of 150 stale acceptances, the 32 with a withdrawn record all found by the subject check; v3.2: 0 of 150, 43 reverted, 21 of them after another hospital's withdrawal) and local EVM; other gaps: local unit tests only
Aggregation integrity (A12), real devices (L4), environment comparison in L1	not evaluated

S2. Trust Assumptions

Table S2 lists the assumptions summarized in Section III-C of the main text.

TABLE S2
Trust assumptions (main text, Section III-C).

	Assumption	If violated
A1	Collision-resistant hash; unforgeable signatures	all bindings void
A2	Ledger consensus preserves order and immutability	non-repudiation lost
A3	Device software does not leak data and checks models correctly	L4 lost
A4	Platform does not authorize uploads for withdrawn subjects	laundering undetected
A5	Auditor is honest and can run a controlled environment in the jurisdiction	L1 void
A6	Local training and jobs are deterministic in the specified stack	L1 false alarms or room for forgery
A7	Audit samples are drawn after commitments and are unpredictable	sample evasion
A8	Withdrawn records are retained encrypted until L3 completes	L3 impossible
A9	Platform does not refuse or delay fee delegation	registration depends on platform
A10	Public remote procedure call (RPC) endpoint is available and reads reflect a confirmed block	stale device checks
A11	Data centers declare r_0 when a withdrawal of level ≥ 2 takes effect	models in scope stay deployable
A12	The aggregator aggregates exactly the accepted updates	rejected updates reach the model undetected

S3. Verification Algorithms

Algorithm S1 gives the acceptance check of Section III-D of the main text.

Algorithm S1 AcceptUpdate$(k, r, \Delta_{k,r}, C_{k,r}, U_{k,r})$

1: check signatures, round, h_{r-1}, $\mathsf{H}(\Delta_{k,r})$; else **reject**(binding)
2: **if** past round deadline **then reject**(late)
3: read withdrawals at a confirmed block; $v_c \leftarrow$ number of effective withdrawals read
4: $Q_k(v_c) \leftarrow$ all records held by k in the withdrawal lists of the first v_c effective withdrawals
5: **for** $q \in Q_k(v_c)$ **do**
6: **if not** VerifyNonMember$(root_S, q, \rho_q)$ **then reject**(policy conflict)
7: **end for**
8: post the decision for (k, r); the contract binds it to the stored $C_{k,r}$ and records τ and the version v_τ at inclusion

Algorithms S2–S5 give the data-center and auditor procedures referred to in Sections III-D and III-E of the main text. They follow the released code; the positive-control rule of Algorithm S5 (line 8) differs from the code that ran the

TABLE S3
CONDITIONAL VERIFICATION CLAIMS (SECTION III-E OF THE MAIN TEXT; GOALS IN SECTION III-C): ASSUMPTIONS, MECHANISMS (OFF-CHAIN ALGORITHM; CONTRACT CHECK), AND COVERAGE IN THIS WORK.

Claim	Assumptions	Algorithm	Contract check	Implemented / evaluated
Enforcement safety	A1, A2, A4–A7, A12	S1, S2, S3	input/update commitment order; acceptance recorded, not checked (v3.2: v_c checked at execution)	yes / E1, E3 (T2, T3, T10); race: measured on Kairos (v3.1 detected, v3.2 reverted)
Request non-repudiation	A1, A2, A9, A10	–	policy version only (v3.2: subject signature, one withdrawal per version)	yes / E1, E3 (T1); authorship (platform-registered key), replay: v3.2, unit tests
Evidence integrity	A1, A2	–	ledger	yes / E3 (T9)
Job binding	A1, A2, A4–A6, L1 premises	S3, S4	approved $(method, prog)$ for jobs; configuration committed before any withdrawal	yes / E3 (T4, T6, T7, T11)
Scope enforcement	A1, A2, A11	–	regular models from declared r_0 not deployable while unresolved; revocation by platform (v3.2: undeclared r_0 as round 1, resolved ranges kept)	partly / E3 (honest r_0 declaration); other gaps (Section III-D): v3.2, unit tests
Deployment consistency	A1–A3, A10	–	`deployed`, `deployable` read	simulated device / E3 (T8, delivery only)
Auditability	A1, A2	–	events; round units keyed by B, job results by h_{out}	yes / E3

experiments, as documented in Section S6. The contract events are listed in the extended results (Table R18).

Algorithm S2 LocalRound(k, r) at data center k

$\sigma \leftarrow$ current policy snapshot (withdrawal version read at a confirmed block)
$S \leftarrow \{x \in D_k : valid(x, \text{now}, \pi)\}$; $root_S \leftarrow$ sorted-Merkle root of $\{rid(x) : x \in S\}$
post $C_{k,r} = (k, r, root_S, |S|, \sigma, \pi)$ ▷ `commitInput`
$seed_{k,r} \leftarrow seed_rule(\text{base seed}, k, r)$ ▷ committed before round 1
$M \leftarrow prog$ run from M_{r-1} on S; batches are a function of $(S, seed_{k,r})$ only
post $U_{k,r} = (k, r, \mathsf{H}(\Delta_{k,r}), h_{r-1}, \mathsf{H}(C_{k,r}))$ ▷ `commitUpdate`
send $\Delta_{k,r}$ to the aggregator

Algorithm S3 ReExecute(k, r): L1 for a sampled unit

$\eta_r \leftarrow \mathsf{H}(\text{blockhash}, r, \text{contract})$ ▷ sampling seed from `sampleAudit`, after aggregation
if $\mathsf{H}(\eta_r \,\|\, k)/2^{256} \geq p$ **then return** not sampled ▷ independent per unit
$S' \leftarrow$ input set rebuilt from the store under the auditor's current policy (a stale snapshot therefore fails the root check below)
if $\text{root}(S') \neq root_S$ **then return** mismatch (input)
if $\mathsf{H}(M_{r-1}) \neq h_{start}$ **then return** mismatch (start state)
if environment $\neq prog$ (image, device, settings, threads) **then return** inconclusive ▷ not implemented; our auditor ran in $prog$
$M' \leftarrow prog$ run from M_{r-1} on S' with $seed_{k,r}$
return pass **if** $\mathsf{H}(M')$ equals the committed $\mathsf{H}(\Delta_{k,r})$ **else** mismatch (output); record against $B(k, r)$

Algorithm S4 InputSet(J): L2 for remediation job J

for each input commitment $C^J_{k,r}$ of J **do**
  $E \leftarrow \{x \in D_k : valid(x, \tau_J, \pi)\}$
  if $\text{root}(E) \neq root_{J,k,r}$ **then** mark (k, r) mismatch ▷ inclusion or omission
  for each withdrawn record q of the subjects covered by J **do**
    if not VerifyNonMember$(root_{J,k,r}, q, \rho_q)$ **then** mark (k, r) included
  end for
end for
return pass if nothing is marked, else mismatch with the marked units

Algorithm S5 ResidualTest$(M, kind)$: L3 with positive control

$a \leftarrow$ offline LiRA AUC of M on (X, Z) with 16 shadows; $se_a \leftarrow$ half-width of the 95% percentile bootstrap CI over 1.96
$\mu, s \leftarrow$ mean and s.d. of the same AUC over n retrained (U0) reference models, excluding M itself if it is one of them
$d \leftarrow a - \mu$; $se \leftarrow \sqrt{se_a^2 + s^2/n}$; $[l, u] \leftarrow d \pm z_{0.95}\, se$
if $-\delta_{\text{L3}} < l$ **and** $u < \delta_{\text{L3}}$ **then** $c \leftarrow$ pass ▷ TOST: both one-sided 5% tests reject
else if $l > 0$ **then** $c \leftarrow$ exceeds ▷ $l > 0$ and $u \geq \delta_{\text{L3}}$: above the retraining mean at the one-sided 5% level and not within the margin; $l > 0$ alone also holds for a CI inside the margin (pass)
else $c \leftarrow$ inconclusive (over-removal if $u < 0$, else insufficient precision) ▷ over-removal: $u < 0$, $l \leq -\delta_{\text{L3}}$; insufficient precision: $l \leq 0 \leq u$, not within the margin
$c_0 \leftarrow$ class of the same test applied to P1, the blocking-only model being replaced (positive control)
if $kind =$ approximate **and** $c =$ pass **and** $c_0 \neq$ exceeds **then** $c \leftarrow$ inconclusive (insufficient power) ▷ control fails when P1 is pass, over-removal, or insufficient precision; the run-time code required only P1's $l > 0$ (Section S6)
return c ▷ deployment requires pass only for approximate methods

Notes on L1 and L3 (main text, Section III-E). Every verification result for a training unit refers to the binding value $B(k, r) = \mathsf{H}(k \,\|\, r \,\|\, h_{start} \,\|\, root_S \,\|\, \sigma \,\|\, \pi \,\|\, prog \,\|\, seed_rule \,\|\, \mathsf{H}(\Delta_{k,r}))$, where $prog$ identifies the program and environment. The four L1 premises are: *batch determinism*, batches are a function of the committed input set and seed rule; a *configuration commitment* $\mathsf{H}(b, seed_rule, prog, \text{normalization scope})$ for base seed b, posted before the first round and any withdrawal, which prevents seed grinding; *set equivalence*, L2 compares the input with the expected set; and *method approval*, because a bit-exact match of an honestly declared fine-tuning program proves nothing about removal. The environment check of Algorithm S3, line 6, is not implemented. For L3, over-removal means only that the CI of the tested model's AUC gap lies below 0 and outside the margin ($u < 0$, $l \leq -\delta_{\text{L3}}$). A pass does not establish the absence of individual influence, an exceeds or inconclusive outcome does not show that unlearning failed, and the positive control shows only that P1's full residual is detectable, not a partial one. δ_{L3} was set on a pilot run (one random 5% cohort, five seeds, not the experimental cohorts) after seeing which methods passed there; it is an exploratory test setting, not a clinical or legal

privacy threshold.

Statistical design. The sample-size procedure (main text, Section III-F) targeted the paired difference tests of test AUROC and LiRA AUC against retraining: minimum effects of 0.005 AUROC and 0.05 AUC, $\alpha = 0.05$ with Holm correction over 18 comparisons, power 0.8, and paired standard deviations of 0.0025 and 0.0156 from five retrained seeds, giving $n = 4$ and $n = 2$; $n = 10$ was chosen as the smallest n at which an exact Wilcoxon test can reach $\alpha/18$. The L3 equivalence test was not sized in advance. A post hoc analysis under the main setting (extended results, Section R10) shows that at 1% withdrawal no model could pass (interval half-width above the margin in every run); elsewhere a method identical to retraining passes with probability 0.54–1.00, the positive control detects residuals of at least 0.04–0.05 AUC with probability 0.8, and, because the interval omits the tested model's training variability, a method at the margin passes with probability up to 0.11 (nominal 0.05; about 0.002 for a fixed model). Outcomes also depend on the margin and interval construction (Table S9); intervals at the nominal level leave the qualitative conclusions unchanged (Table S10; scenario-level t interval over the ten seeds: the control detects in 9 of 19 scenarios, none early or mid, a statement about a method within a scenario rather than an audit decision on one model).

S4. Conditional Verification Claims

Table S3 summarizes the conditional verification claims of Section III-E of the main text; Table S1 (Section S1) maps each conclusion to its evidence.

Under the assumptions of Table S2, a duty (main text, Section III-B) is verifiably complete when:

- S_1: every input commitment accepted after t_{eff} carries a valid non-membership proof for each of the patient's records; job inputs, whose commitments are kept off the ledger, are checked by the auditor's set comparison instead;
- S_2: every model in the patient's influence scope is revoked, and the deployed model is outside the scope or is the output of a verified job;
- S_3: the job's output has passing L1 and L2 results recorded by the auditor under its output hash and, if the method is approximate, a passing L3 result with a positive control.

Completion is conditional: re-execution covers only sampled units, and L3 is a statistical decision rule, not a proof that influence is absent.

S5. Additional Results

Tables S4–S5: E2 per scenario; Table S6: concealment cases T11a–f by verifier. Per-condition and supporting tables are in the extended results: E0 reproducibility and the multi-container run (Tables R4 and R14), E1 per condition and on Kairos (Tables R5–R6), the auxiliary conditions (Tables R8–R9), the L1 sampling detection probability (Fig. R1), the evidence-component ablation (Table R12), ledger costs (Tables R13–R15), and the implementation notes, hyperparameters, and fault-injection design (Section R3). Each E3 case (main text, Table III) escapes detection when: T1, the delay length is not provable; T2a, T2b, T3, the unit is not sampled (T2a and T3 also when the non-membership tree is malformed); T4, T7, the job unit is not sampled ($p_J < 1$); T5, the residual is below the test's sensitivity; T6, records are hidden from the store; T8, the device software is tampered with or the v3.1 deployability gap is exploited; T9, consensus fails; T10, the auditor colludes or the unit is not sampled.

TABLE S6
E3, concealment cases T11a–f: detection by verifier (20 injections per variant, five ledger runs). a: fine-tuning until L3 passes; b: honest retraining with the withdrawers re-injected in the last round; c: forged trajectory (Adv-F of Zhang et al.); d: neighbor substitution (Adv-R); e: seed grinding; f: unapproved program. Final $\varepsilon = 0.65$: relative distance between two honest nondeterministic retrainings at seed 1 (0.646 between P1 and U0); ε_r: per-round tolerance with batches from the committed seed rule; submitted-batch: per-round replay of the submitted batches at any $\varepsilon_r \geq 0$.

Variant	L3 only	final ε	$\varepsilon_r \leq 10^{-3}$	$\varepsilon_r = 10^{-2}$	submitted-batch	B4
a	0/20	16/20	–	–	–	20/20
b	0/20	0/20	20/20	7/20	–	20/20
c	20/20	16/20	20/20	20/20	20/20	20/20
d	4/20	4/20	20/20	20/20	0/20	20/20
e	0/20	6/20	–	–	–	20/20
f	20/20	16/20	–	–	–	20/20

S6. Rule Provenance of the L3 Positive Control

TABLE S7
Positive-control rule behind each reported L3 item and whether it differs from the run logs (run-time rule, $l > 0$).

Reported item	Source	Rule applied	Differs from run logs
Main text, Table II (base scenario)	10 run logs	as written	no
Tables S4–S5 (19 scenarios)	190 run logs	as written	53 fields (Table S8); U0 self-reference
Main text, Section IV-B: passes 57→0, 58→2, 53–60→3	80 run logs	before: TOST pass, no control; after: as written	both recomputed
Extended results, Table R7	190 run logs	each variant, control re-applied	recomputed
Extended results, Tables R8–R9	auxiliary logs	as written	per wave 4 of 16; others none
Main text, Table III (T5) and the deployable state of 515 jobs	5 ledger runs	L3 without control; all but T5 exact	no

The positive control of Algorithm S5 (line 8) credits an approximate method's pass only if the blocking-only model P1 is classified *exceeds* under the same test. The code that ran the experiments (tag `final-rerun-v1`, results committed 2026-10-30) credited the pass whenever P1's 90% CI lower bound was above 0, which also admits runs in which P1 itself passed; the rule as written was restored in the released code on 2026-11-01, after the runs. E2 did not use the ledger, so the affected records are the `deploy` fields of the run-log files. Every L3 outcome and deployment count in the manuscript is recomputed from the stored statistics under the rule as written (`scripts/summarize_e2.py`); Table S7 lists the rule behind each reported item. Re-applying the rule

TABLE S4

E2 PER SCENARIO, MEAN OVER 10 SEEDS: TEST MACRO AUROC DIFFERENCE FROM U0 ($\times 10^{-3}$) AND LIRA AUC DIFFERENCE FROM U0 ($\times 10^{-2}$). LATE SCENARIOS: THE FOUR LATE RATE ROWS, MI, WOMEN ≥ 65, ONE INSTITUTION, THREE WAVES, INSTITUTION EXIT, LATE-JOINING; RE-CONSENT COUNTS AS EARLY/MID IN FIG. 3 OF THE MAIN TEXT.

	P1		U1		U2		U3		U4		U4s	
Scenario	AUROC	LiRA	AUROC	LiRA	AUROC	LiRA	AUROC	LiRA	AUROC	LiRA	AUROC	LiRA
5%, late (base)	+0.2	+6.3	-0.3	+0.1	-10.6	-0.8	-8.7	-1.2	+0.3	+6.4	+0.3	+5.6
1%, late	-0.1	+9.8	-0.2	+1.3	-11.1	+3.8	-8.4	+2.6	-0.1	+9.9	-0.0	+9.5
10%, late	+0.6	+6.6	-0.4	+0.4	-10.7	-0.9	-8.7	-0.2	+0.6	+6.4	+0.5	+4.9
20%, late	+2.8	+6.5	-0.3	+0.2	-9.4	-1.8	-8.3	-1.2	+2.8	+6.3	+2.6	+4.4
5%, early	+0.1	+0.6	+0.4	+0.5	-10.6	-2.3	-8.6	-1.7	+0.2	+0.8	+0.2	+0.1
5%, mid	+0.4	+1.9	+0.3	-0.2	-10.4	-0.9	-8.6	-0.3	+0.4	+1.7	+0.4	+1.1
1%, early	-0.6	+0.2	-0.3	+1.3	-10.8	-3.1	-8.7	-0.6	-0.5	+0.3	-0.5	+0.0
1%, mid	+0.2	+2.1	-0.4	-0.5	-10.2	-3.7	-8.6	-5.1	+0.3	+1.1	+0.3	-0.8
10%, early	-0.0	-0.4	-0.2	+0.0	-10.4	-1.1	-7.9	-0.8	+0.1	-0.6	-0.3	-1.2
10%, mid	+1.0	+2.0	+0.4	+0.1	-10.7	-2.0	-8.1	-1.8	+1.1	+1.9	+1.2	+1.3
20%, early	+0.8	-0.5	+0.1	-0.4	-11.4	-1.2	-8.9	-0.5	+0.8	-0.7	+0.6	-1.4
20%, mid	+1.2	+1.5	+0.5	-0.6	-10.9	-1.0	-9.0	-0.6	+1.2	+1.4	+1.0	+0.1
MI patients	+0.6	+9.3	-0.1	+0.1	-10.4	+1.5	-8.4	+2.3	+0.6	+9.1	+0.5	+7.4
women aged ≥ 65	+0.4	+6.4	-0.2	-0.6	-10.1	-1.3	-8.0	-1.0	+0.4	+6.5	+0.4	+5.5
one institution	+1.0	+6.1	-0.1	+0.2	-9.7	-3.7	-8.4	-1.9	+1.0	+6.0	+1.0	+5.4
three waves	+0.7	+3.5	-0.1	+0.3	-10.6	-1.0	-8.1	-0.2	+0.7	+3.4	+0.7	+2.7
re-consent	-0.0	+1.1	-0.2	-1.9	-10.4	-3.1	-8.5	-3.4	-0.1	+1.0	-0.0	+1.3
institution exit	+2.6	+4.8	-0.2	+0.2	-8.6	-4.3	-7.8	-2.5	+2.6	+4.5	+2.3	+1.9
late-joining cohort	+0.6	+6.8	+0.0	+0.0	-10.8	-0.1	-8.3	+0.0	+0.6	+6.6	+0.6	+5.6

TABLE S5

E2 PER SCENARIO: L3 OUTCOMES AS PASS / EXCEEDS / INCONCLUSIVE / OVER-REMOVAL, COUNTED OVER SEEDS. INCONCLUSIVE INCLUDES A FAILED POSITIVE CONTROL.

Scenario	P1	U0	U1	U2	U3	U4	U4s
5%, late (base)	1/9/0/0	10/0/0/0	9/0/0/1	9/0/1/0	9/0/1/0	0/10/0/0	0/9/1/0
1%, late	0/9/1/0	0/1/8/1	0/1/9/0	0/0/10/0	0/0/10/0	0/8/2/0	0/10/0/0
10%, late	0/10/0/0	10/0/0/0	10/0/0/0	10/0/0/0	10/0/0/0	0/10/0/0	0/10/0/0
20%, late	0/10/0/0	10/0/0/0	10/0/0/0	10/0/0/0	10/0/0/0	0/10/0/0	1/9/0/0
5%, early	10/0/0/0	8/0/0/2	0/0/10/0	0/0/6/4	0/0/9/1	0/0/10/0	0/0/10/0
5%, mid	8/2/0/0	8/1/0/1	2/0/8/0	2/0/8/0	2/0/8/0	0/2/8/0	1/1/8/0
1%, early	0/1/9/0	0/1/9/0	0/0/9/1	0/0/10/0	0/0/10/0	0/1/9/0	0/0/10/0
1%, mid	0/0/10/0	0/1/9/0	0/0/9/1	0/0/7/3	0/0/3/7	0/0/10/0	0/0/10/0
10%, early	10/0/0/0	10/0/0/0	0/0/10/0	0/0/10/0	0/0/10/0	0/0/10/0	0/0/9/1
10%, mid	9/1/0/0	10/0/0/0	1/0/9/0	1/0/9/0	1/0/9/0	0/1/9/0	1/0/9/0
20%, early	10/0/0/0	10/0/0/0	0/0/10/0	0/0/10/0	0/0/10/0	0/0/10/0	0/0/10/0
20%, mid	10/0/0/0	10/0/0/0	0/0/10/0	0/0/10/0	0/0/10/0	0/0/10/0	0/0/10/0
MI patients	0/10/0/0	5/1/2/2	9/0/1/0	8/0/2/0	2/1/7/0	0/10/0/0	0/10/0/0
women aged ≥ 65	0/10/0/0	9/1/0/0	6/1/3/0	10/0/0/0	10/0/0/0	0/10/0/0	0/10/0/0
one institution	1/9/0/0	8/0/0/2	9/0/0/1	0/0/0/10	6/0/1/3	0/10/0/0	0/9/1/0
three waves	4/6/0/0	10/0/0/0	6/0/3/1	6/0/4/0	6/0/4/0	1/6/3/0	2/4/4/0
re-consent	7/1/2/0	8/0/2/0	0/0/7/3	0/0/7/3	0/0/5/5	0/1/9/0	0/1/9/0
institution exit	4/6/0/0	10/0/0/0	6/0/4/0	0/0/0/10	2/0/2/6	2/6/2/0	6/0/4/0
late-joining cohort	0/10/0/0	9/0/0/1	9/0/0/1	10/0/0/0	10/0/0/0	0/10/0/0	0/10/0/0

TABLE S8

APPROXIMATE-METHOD L3 PASSES UNDER NO CONTROL, THE RUN-TIME RULE (P1'S $l > 0$), AND THE RULE AS WRITTEN (P1 *exceeds*). TOP: THE 80 EARLY- AND MID-WITHDRAWAL RUNS. BOTTOM: THE FOUR SCENARIOS (10 RUNS EACH) WHOSE COUNTS DIFFER, RUN-TIME $\rightarrow$ AS WRITTEN; ALL OTHER SCENARIOS ARE IDENTICAL.

	U1	U2	U3	U4	U4s
No control	60	53	59	57	58
Run-time rule ($l > 0$)	11	11	11	8	10
Rule as written (exceeds)	3	3	3	0	2
20%, mid	5→0	5→0	5→0	5→0	5→0
10%, mid	4→1	4→1	4→1	3→0	4→1
institution exit	9→6	0	4→2	3→2	9→6
three waves	7→6	7→6	7→6	1	3→2

TABLE S9

L3 PASSES OVER ALL 190 RUNS UNDER FOUR SETTINGS (z OR t CRITICAL VALUE, $n_{\text{ref}} - 1$ D.F.), POSITIVE CONTROL RE-APPLIED AS WRITTEN; FROM EXTENDED RESULTS TABLE R7.

Setting	U0	U1	U2	U3	U4	U4s
$\delta_{\text{L3}} = 0.05$, z (main)	145	77	66	68	3	11
$\delta_{\text{L3}} = 0.025$, z	40	23	5	23	0	7
$\delta_{\text{L3}} = 0.05$, t	122	61	51	54	2	8
$\delta_{\text{L3}} = 0.025$, t	14	6	1	5	0	4

(`scripts/recount_l3_deploy.py`) changes 53 of the 269 run-log fields recorded as deployable on an L3 pass to inconclusive (U1 12, U2 9, U3 11, U4 9, U4s 12), all in the four scenarios of Table S8 in which P1 passed TOST with a lower bound above 0; the base scenario has no changed field, the per-wave results of the three-wave scenario change in 4 of 16 fields, and $K = 5$, lead I, and Challenge 2021 in none of 45. The C3 ledger runs are unaffected: every job there but T5 declared retraining (exact), deployable on L1 and L2 alone; T5 declared FedOSD (approximate) and was always *exceeds*, which the control does not alter.

The decision of 2026-10-27 that introduced the control glossed *exceeds* as "lower bound above 0", and the run-time code implemented that gloss, whereas the classification in the

same code (Algorithm S5, line 5) defines *exceeds* as "not within the margin and lower bound above 0". Under the gloss a pass was credited even when P1 itself passed TOST, the case the control was introduced to exclude: in seed 1 of the 20% mid-training scenario P1's CI was $(0.006, 0.033)$, above 0 but inside the margin; in seed 1 of the base scenario it was $(0.043, 0.087)$, outside the margin, and both rules credited the control. The code review of 2026-11-01 aligned the control with the classification; because the alignment came after the runs, Table S8 reports the approximate-method passes in the 80 early- and mid-withdrawal runs under no control, the run-time rule, and the rule as written. No qualitative finding depends on the choice: approximate methods passed in at most 11 of the 80 runs under the run-time rule and 3 under the rule as written, the base scenario and retraining (exact; control not applied) are unchanged, and what changes is the number of deployable outputs in four scenarios. `scripts/l3_rule_comparison.py` reproduces Table S8 and the 53 fields from the run logs.

S7. L3 Sensitivity to the Test Setting

TABLE S10
L3 UNDER INTERVALS THAT APPROACH OR HOLD THE NOMINAL LEVEL (POST HOC; MARGIN 0.05 THROUGHOUT; THE RUN-LEVEL RULE IS UNCHANGED). BOUNDARY: LARGEST SIMULATED PROBABILITY OVER THE SCENARIOS THAT A METHOD WHOSE TRUE DIFFERENCE EQUALS THE MARGIN PASSES (NOMINAL 0.05), UNDER A NORMAL MODEL WITH EACH SCENARIO'S REFERENCE STANDARD DEVIATION, ITS MEAN BOOTSTRAP STANDARD ERROR, AND FIVE REFERENCES, WHEN THE METHOD'S TRAINING VARIABILITY EQUALS THAT OF RETRAINING ($\sigma_T = \sigma_{\text{ref}}$) OR IS 1.5 TIMES LARGER (EXTENDED RESULTS, SECTION R10). RUN LEVEL: PASSES OVER THE 190 RUNS WITH THE POSITIVE CONTROL APPLIED UNDER THE SAME INTERVAL, AND RUNS IN WHICH THE CONTROL DETECTED (P1 *exceeds*). SCENARIO LEVEL: FOR EACH OF THE 19 SCENARIOS, THE MEAN OF THE TEN SEEDS' PAIRED LIRA AUC DIFFERENCES FROM RETRAINING WITH A 90% t INTERVAL, CLASSIFIED BY THE SAME RULE; PASSES CREDITED ONLY WHERE P1 IS *exceeds* (WITHOUT THE CONTROL: U1 19, U2 16, U3 18, U4 10, U4S 12 OF 19).

	Boundary		Passes						P1
Interval	σ_{ref}	$1.5\,\sigma_{\text{ref}}$	U0	U1	U2	U3	U4	U4s	exceeds
Run level (of 190)									
main: s^2/n_{ref}, z	0.114	0.200	145	77	66	68	3	11	94
s^2/n_{ref}, t	0.061	0.135	122	61	51	54	2	8	88
prediction: $s^2(1+1/n_{\text{ref}})$, z	0.043	0.104	123	60	50	57	2	8	88
prediction, t	0.017	0.053	83	43	37	36	0	1	77
Scenario level (of 19)									
paired differences, t (9 d.f.)	nominal	nominal	–	9	7	9	0	2	9

Table S9 gives the L3 passes over all 190 runs under the main setting and three stricter ones, each with the positive control re-applied under the rule as written (Section S6); U0 is tested against the other reference models. Pass counts fall under a narrower margin and a t critical value for every method, including exact retraining (U0: 145 to 14), so an L3 outcome is conditional on the margin and interval construction, and an inconclusive outcome under any setting does not establish that remediation failed. The qualitative findings held under every setting: under the t rule the blocking-only model still exceeded in 9/10 base-scenario runs, FedEraser still showed over-removal in every institution-concentrated and institution-exit run, and at most two passes per approximate method remained after the positive control in the 80 early- and mid-withdrawal runs. Table S10 repeats the count under three interval constructions that approach or hold the nominal level; the run-level rule, fixed before the reported runs, is kept for run-level decisions. Agreement of the qualitative findings across constructions is not nominal error control: under the simulation model of Table S10, the run-level rule passes a method whose true difference equals the margin with probability up to 0.114 when the method varies as retraining does and 0.200 when it varies 1.5 times more, whereas the prediction-interval and scenario-level constructions hold the level only under their stated assumptions. The number of shadow models is unlikely to explain the missing discrimination: scoring each stored shadow model with 4, 8, or 15 of the others gave attack AUCs of 0.503, 0.503, and 0.503 after early, 0.513, 0.516, and 0.517 after mid-training, and 0.549, 0.558, and 0.562 after late withdrawal (more than 16 were not trained).

S8. Comparison with Prior Work and Contract Gaps

Comparison with prior work (main text, Sections I–II). The extended results (Table R16) compare twelve works along the features this work targets. Seven record consent or remediation on a ledger or shared log: Zhuang et al. (2020; access lists for health information exchange), Khan et al. (2025; update records), Kostick-Quenet et al. (2025; consent filter at training, proposed), Fdhila et al. (2022; commitments and auditor re-execution, non-retroactive), VeriFUL (Nguyen et al., 2026; signed requests, membership-inference threshold, concept), Zuo et al. (2025; request and action records, forgotten-class accuracy), and Kasiraju et al. (2025; lineage hashes, periodic checks). None reports a test on a public ledger of whether a missed withdrawal is detected, decides consent at update acceptance, or combines execution evidence with a positive-controlled residual test; the five verifiable-training or verifiable-unlearning works (TrustFL, Li et al., Weng et al., Eisenhofer et al., VeriFi) do not record consent. Entries describe what each work set out to do, not whether it failed; references are listed with the table.

Contract gaps (main text, Section III-D). Beyond the acceptance race, ConsentLedger v3.1 (all experiments) has these gaps (extended results, Table R2): unauthenticated, replayable withdrawals; re-consent registered by the platform alone; an undeclared or mis-declared r_0 leaving in-scope models deployable; scope released after resolution; reusable round numbers and model hashes; a stallable round; and a non-membership verifier that trusts the prover's tree. Contract v3.2 rejects an acceptance whose v_c differs from the withdrawal version at execution (measured on Kairos, Table S1) and addresses the others under local unit tests only; seven fault paths that succeed against v3.1 are rejected (replayed withdrawal, unsigned re-consent, acceptance after a same-block withdrawal, undeclared r_0, late r_0 from another data center, round reuse, scope after resolution). Four paths remain open, among them count-based matching of job outputs to withdrawals, forgery by the platform that registers the subject key, and a stalled round. E1–E4 measured v3.1.